SMARTBRIDGE
Let's Bridge the Gap

SKILL WALLET
A SmartBridge Product

Project Executable Files

Date	29 Jun 2026
Team ID	xxxxxx
Project Name	OptiCrop
Maximum Marks	3 Marks

Step 1: Submission Checklist

S.No	Item to Submit	Submitted (Yes / No / NA)
1	Complete source code (all files and folders)	Yes
2	README / Setup Guide	Yes
3	requirements.txt / dependency file	Yes
4	Database schema / seed files	Yes
5	Environment configuration file (.env.example or similar)	Yes
6	Deployed application URL	Yes
7	APK / Executable binary	NA
8	Dockerfile / Containerization config	NA
9	Test files and test results	Yes
10	Demo video or walkthrough	No

Step 2: File / Folder Structure

OptiCrop/

│

```
├── static/              - CSS, JavaScript, Images
├── templates/             - HTML templates (Frontend pages)
├── venv/                - Python virtual environment (not uploaded to GitHub)
├── app.py               - Main Flask application
├── train_model.py          - Machine learning model training script
├── crop_model.pkl          - Trained machine learning model
├── Crop_recommendation.csv - Crop recommendation dataset
├── database.py            - Database-related operations
├── requirements.txt        - Project dependencies
├── README.md              - Project setup and execution guide
└── .gitignore             - Git ignore configuration
```

Step 3: Deployment / Access Details

Field	Details
Hosted / Deployed URL	https://opticrop-m04h.onrender.com
Login Credentials (Demo)	Mail:daganisuryaprakash@gmail.com Password:Surya@1234 pass
Platform / Hosting Provider	
Repository Link	https://github.com/bhagathshankar37/opticrop
Demo Video Link	

Step 4: Run Instructions

Pre-requisites

- Python 3.10 or above
- Flask
- NumPy
- Pandas
- Scikit-learn
- Joblib

- Web browser (Chrome, Edge, or Firefox)
- Git (optional, for cloning the repository)

Step 1

Clone the repository:

git clone <GitHub Repository URL>

Step 2

Navigate to the project directory:

cd OptiCrop

Step 3

Install the required dependencies:

pip install -r requirements.txt

Step 4

Run the Flask application:

python app.py

Step 5

Open the URL displayed in the terminal (typically http://127.0.0.1:5000/) in your web browser.

Step 6

Enter the required soil and environmental parameters:

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- Temperature
- Humidity
- pH
- Rainfall

Click **Predict** to view the recommended crop.

Step 5: Known Issues / Limitations

S.No	Known Issue / Limitation	Workaround / Status
1	The application runs only on a local Flask server if not deployed.	Can be deployed on Render or another cloud platform.
2	Prediction accuracy depends on the quality and size of the training dataset.	Use a larger and more diverse dataset to improve accuracy.
3	No user authentication or account management is implemented.	Can be added as a future enhancement if required.